David Le Chan

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Education

M.S. in Electrical and Computer Engineering (ECE) Expected May 2027
Carnegie Mellon University (CMU) | Pittsburgh, PA

- GPA: /4.00

B.S. in Electrical and Computer Engineering (ECE) Graduated May 2026
Carnegie Mellon University (CMU) | Pittsburgh, PA

- GPA: 3.8/4.0; *University Honors, 6x College of Engineering Dean's List Recipient*

Work Experience

Hardware Development Engineering Intern May 2026 - August 2026
Annapurna Labs (Amazon AWS), Machine Learning Acceleration | Austin, TX

- Validated and deployed PCIe Gen5/Gen6 connectivity to scale machine learning acceleration systems
- Designed and executed qualification tests for switches, expanding high-speed rack-level test coverage
- Migrated tests to a Lua orchestration framework to enable continuous validation via AWS test infrastructure
- Identified mechanical defects impacting 100k+ units, driving resolution to meet deployment schedules

FPGA & Electrical Engineering Intern May 2025 - August 2025
KLA Corporation, Broadband Plasma Semiconductor Inspection | Milpitas, CA

- Upgraded image compression FPGA firmware, achieving a 6.7% throughput increase on flagship inspection tools
- Designed Verilog pipelines for image segmentation and reconstruction, verifying functionality via Questa
- Validated designs on Alveo PCIe cards, closing timing at 80MHz with 50% more parallel processing engines

Undergraduate Research Assistant May 2024 - Present
IO Harness Project, CMU ECE | Pittsburgh, PA

- Designing a standardized research-chip harness to minimize infrastructure rework; testing a 180nm prototype
- Authored and verified SystemVerilog RTL for UART, I2C, and SPI communication blocks
- Streamlining an eFPGA-generation pipeline to serve as a device under test (DUT)

Power Electronics & Programming Intern May - August 2023
Tau Motors | Redwood City, CA

- Independently designed and prototyped motor testbenches and PCBs, owning layout, assembly, and validation
- Built Python inventory management tools from scratch to accelerate hardware testing cycles

Leadership and Projects

Hybrid FFT/NTT Accelerator SoC January 2026 - Present

- End-to-end design of a novel hybrid accelerator that supports both FFT and NTT operations through hardawre reuse, culminating in a 28nm tapeout
- Integrated our hybrid accelerator with a RISC-V core to create a test SoC in SpinalHDL
- Designed a custom cocotb-based test orchestrator for multi-level verification at all design stages

Head Teaching Assistant (TA), Introduction to ECE (18-100) January 2025 - January 2026

- Led a 40 TA team to foster an emotionally-safe environment where 180 first-year students can explore ECE, form lasting friendships, and develop strong engineering habits
- Continuously redesigned lab curriculum to give students hands-on experience with real-world design workflows
- Established automated Python scripts and feedback systems to streamline course operations, reducing administrative overhead and empowering TAs to focus on mentorship and teaching quality